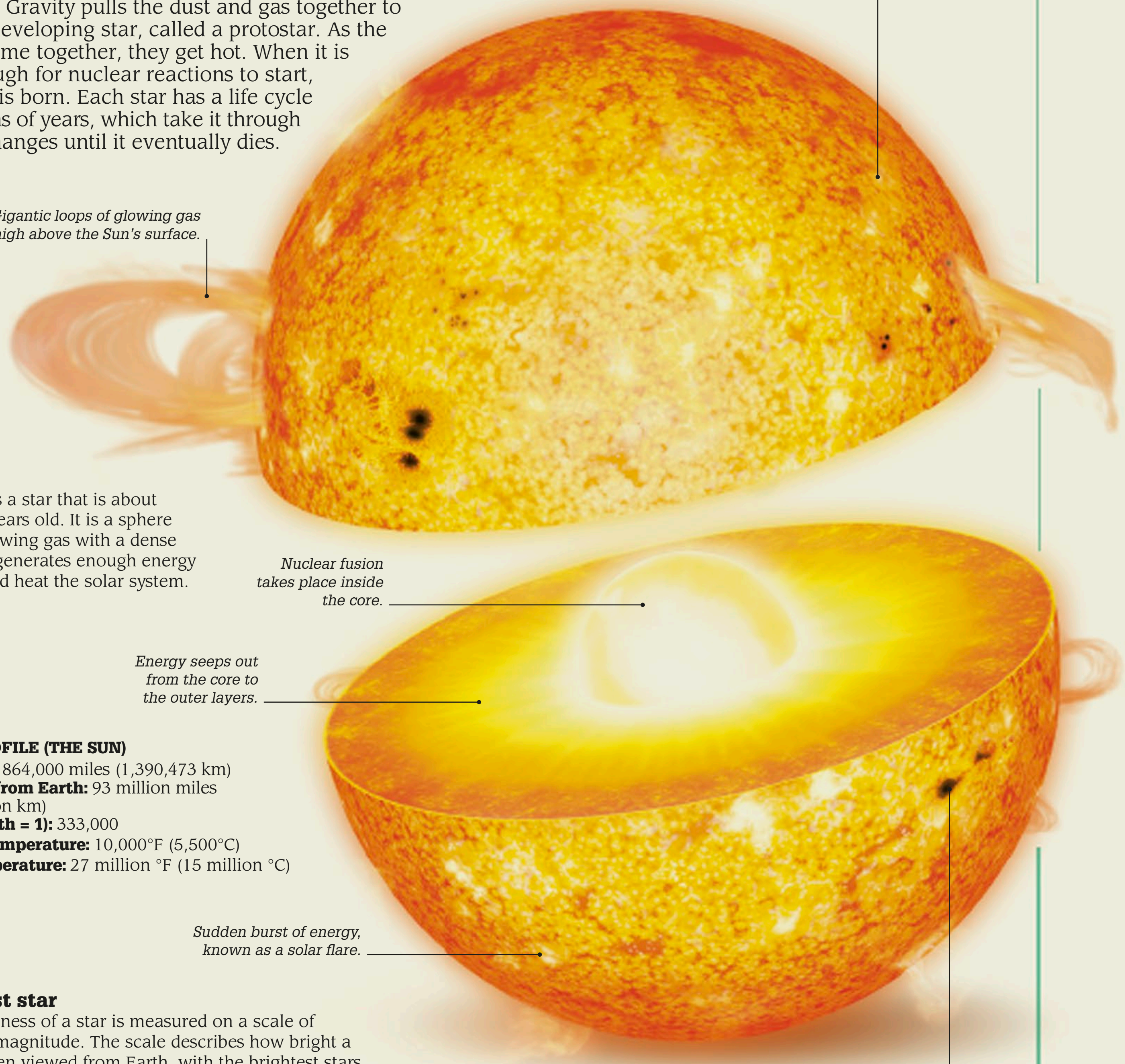
Stars

A star is a brilliant, shining ball of extremely hot gas that generates huge amounts of energy in its core. Stars are created from clouds of gas and dust, known as nebulas. Gravity pulls the dust and gas together to form a developing star, called a protostar. As the gases come together, they get hot. When it is hot enough for nuclear reactions to start, the star is born. Each star has a life cycle of billions of years, which take it through many changes until it eventually dies.



The Sun

The Sun is a star that is about 5 billion years old. It is a sphere of hot, glowing gas with a dense core that generates enough energy to light and heat the solar system.

DATA PROFILE (THE SUN)
Diameter: 864,000 miles (1,390,473 km)
Distance from Earth: 93 million miles (150 million km)
Mass (Earth = 1): 333,000
Surface temperature: 10,000°F (5,500°C)
Core temperature: 27 million °F (15 million °C)

Brightest star

The brightness of a star is measured on a scale of apparent magnitude. The scale describes how bright a star is when viewed from Earth, with the brightest stars having the lowest numbers. The Sun, with a magnitude of –26.74, is the brightest object in our skies.

Apparent magnitude